Mouhssine El Boumshouli

AI Engineering Student | Full-Stack & AI Software Developer | Founder, Convoroute LLC
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Born 28 November 2004, Fès, Morocco | Moroccan national

Profile

Fourth-year AI engineering student at EIDIA, Euromed University of Fes (graduating 2028), and founder of Convoroute LLC; I build full-stack products and reproducible AI research.

Education

EIDIA, Euromed University of Fes (UEMF) — Fès, Morocco
Engineering Degree in Artificial Intelligence, 5-year programme — *2023 – Expected 2028*

- Currently in the 4th year of the programme, which is the 2nd year of the engineering cycle; the integrated preparatory cycle (CPI) is completed.
- **Relevant coursework:** supervised machine learning, unsupervised machine learning, multilayer neural networks, image processing, computer graphics, SQL and NoSQL databases, data analysis in R, advanced algorithms and data structures, probability and statistics, operations research.

Institut Mazarin — Fès, Morocco
Baccalauréat, Physical Sciences, French option (international track) — *June 2023*

Research Experience

DARE-Bench – AI Agent Reliability Study — 2026

- A study of one question: do extra turns make an AI agent more reliable, or just more random?
- Raised measured AI-agent success from **34.2% to 50.0%** (+15.8 points) by giving the agent 5 interaction turns instead of 3, over **240 reproducible runs across 24 tasks**, 5 repetitions per condition.
- Made the finding trustworthy (flakiness also increased) with official rescoring of every run and **10,000-resample bootstrap confidence intervals**. Code published openly (Apache 2.0).

medskel – Medical Image Skeletonization — 2026

- Software that turns vessel-like shapes in medical images into clean centrelines, so vessel length is measured the same way every time.
- Cut disagreement between two human observers on vessel-length measurement from **9.3% to 3.2%** across **28 retinal images** (lower on 26 of 28 images, $p < 0.0001$) by implementing a bisector-based skeletonization method and testing it against standard pixel thinning. Code published openly (MIT).
- Proved it with **28 hand-derived tests and 9 repeatable experiments**, and reported the cost openly: about **66× slower** runtime than thinning.

Professional Experience

Bounaim Auto — Beni Mellal, Morocco
Software & AI Engineering Intern — *July 1 – August 26, 2026*

- A platform for a car-import company: it compares supplier quotations and shows the true total cost of each offer, so the buyer sees the full picture.
- Centralised **5 procurement workflows** (supplier quotes, currency and cost analysis, document imports, AI-assisted data capture, human review) into one platform, SmartImport, built with React, TypeScript, FastAPI, PostgreSQL, and Docker.
- Reached **98.99% document-field F1 and 100% product-line F1** on synthetic evaluation, and **92.31% top-1 supplier matching** on eligible unique-match cases, by combining deterministic landed-cost logic with human-reviewed AI extraction.
- Verified with **139 automated backend tests**; incomplete offers are flagged and every analysis is kept on record.

Convoroute LLC — Remote
Founder & Software Developer — *June 2026 – Present*

- My own company: AI chatbots that plug into a business's website and answer its customers automatically.
- Founded Convoroute LLC and launched a B2B SaaS product covering **3 business needs** (customer support, FAQ automation, lead capture) by building AI chatbots that embed directly on client websites.
- Run the whole product myself – architecture, LLM integration, deployment, onboarding, and Stripe billing – in one repeatable workflow.

Projects

Recall – Multilingual Memory & Transcription App (React Native, Expo, SQLite, Gemini) 2026

- A mobile app that records conversations, classes, and meetings, transcribes them live in several languages, and turns them into clear notes, so nothing important is lost.
- Kept recordings safe through connection drops by storing a local WAV file while streaming audio in roughly **100 ms chunks** for live transcription (sessions rotate about every **8.5 minutes**).
- Turned transcripts into summaries, key points, action items, and chapters, stored in **versioned SQLite** with offline history and playback.

UEMF Présence – Smart Attendance System (Next.js, TypeScript, Prisma) 2026

- A digital attendance system for my university: it replaces paper sign-in sheets in class.
- Reduced false check-ins for **3 user groups** (students, professors, administrators) by verifying GPS location at check-in and refreshing **HMAC-SHA256 QR codes every 10 seconds**.
- Cut routine admin work with recurring schedules, review tools, CSV reports, and an anomaly dashboard.

AutoStock AI – Inventory & Forecasting for Odoo (Python, React) 2026

- An application that connects to Odoo to help a business track its stock and forecast what it will need next.
- Completed and merged the **first 6 development gates** of the application by October 2026, each gate shipped as a small pull request with automated tests.
- Built with an AI coding agent; every change independently reviewed (code, tests, screenshots) before merging.

Startup Internship Hunter (Python, Playwright, SQLite) 2026

- A tool I built for my own internship search: it collects public information about startups and prepares human-checked outreach.
- Built an evidence-first research pipeline with hard safety limits (max **12 pages and 2 repositories per company**, no automated logins or forms) to prepare human-verified internship outreach.

Additional projects 2025 – 2026

- Also built: **Intelligent Student Performance System** (Java, 2025; logistic regression written from scratch); **Modern Finance Tracker** (C++17, Qt, SQLite, 2025); **Jev Voice Assistant** (Python, 2026; text commands routed through a decision-model API); **food-expiry iOS app** (TypeScript, in development).

Technical Skills

- **Languages:** Python, TypeScript, JavaScript, Java, C, C++, SQL.
- **AI & Data:** LLM agents, AI evaluation, OpenAI and Gemini APIs, Google Cloud Speech-to-Text, scikit-learn, pandas, NumPy, OpenCV.
- **Web & Backend:** React, Next.js, React Native, FastAPI, SQLAlchemy, Pydantic, Prisma, PostgreSQL, SQLite.
- **Tools & Practices:** Git/GitHub, Docker, Linux, pytest, Playwright, Vercel, Nginx, Qt, CMake, Stripe, WordPress/WooCommerce, Cloudflare.

Certifications & Languages

- **UEMF Language Center** – C1 English (Cambridge English Placement Test), 26 October 2026.
- **TOEFL iBT** – 94/120 (Reading 20, Listening 22, Speaking 25, Writing 27), December 2022.
- **EF SET English Certificate** – 68/100, CEFR C1, August 2022 | **EnglishScore (British Council)** – CEFR C1, 535, August 2022.
- **Google** – Foundations of UX Design (Coursera) and Digital Marketing certificates, August 2022.
- **Languages:** Arabic (native) | English (C1) | French (B1) | Spanish (beginner). **Activities:** MSIC 2026 Hackathon | Idea'Wave Ideation Programme | 20+ GitHub repositories | Fiverr freelance app development.

Academic References

Pr. Asmae Abadi – Associate Professor at Euromed University of Fes; Coordinator of the Artificial Intelligence Program and the Industry 4.0 Program. Recommendation letter provided (October 2026).

Pr. Loubna Ourabah – Professor of Computer Science and Artificial Intelligence at EIDIA, Euromed University of Fes; Coordinator of the Big Data Analytics Program. Recommendation letter provided (October 2026).

Pr. Ayoub El Baraka – Physics researcher with a PhD from Sidi Mohamed Ben Abdellah University; taught Point Mechanics and MATLAB/Simulink Programming at Euromed University of Fes. Recommendation letter provided.